UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS

FACULTAD DE CIENCIAS FÍSICAS

Teoría General de la Relatividad

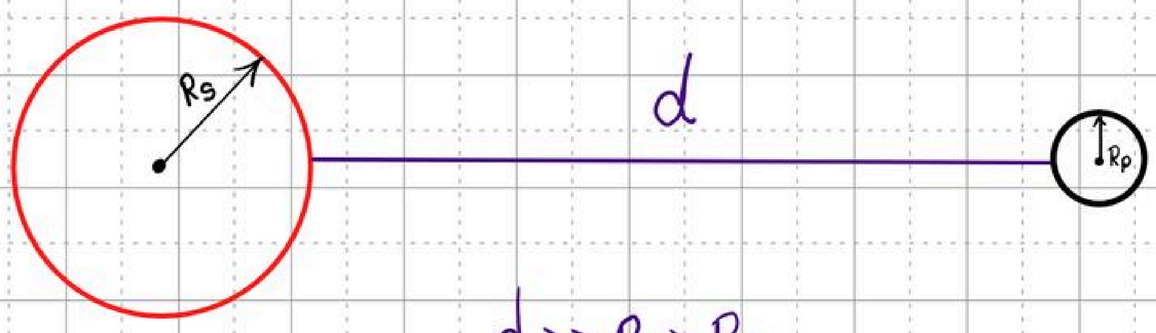
4ta. práctica
(Segundo semestre 2025)

Indicaciones Generales:
Materiales a utilizar: libros y apuntes.
Puntaje total: 20 puntos.
Las soluciones deben ser presentadas en el orden establecido de la práctica.
Fecha de entrega: **07/12/2025**.
Enviar la práctica resuelta al siguiente correo: **richard.avalos@unmsm.edu.pe**

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1. **(2p)** Un planeta orbita a una estrella de forma tal que la distancia entre la estrella y el planeta es mucho mayor a sus radios. Derive la ecuación de movimiento relativista del planeta.



$d >> R_S > R_P$

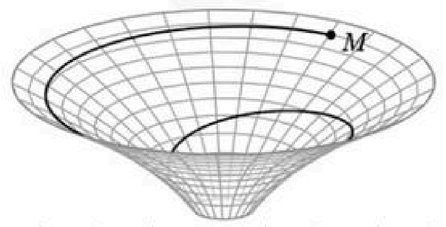
Solución de Schwarzschild

$$ds^2 = -\left(1-\frac{R_s}{r}\right)c^2dt^2 + \frac{1}{\left(1-\frac{R_s}{r}\right)}dr^2 + r^2(d\theta^2 + \operatorname{sen}^2\theta\, d\phi^2)$$



Métrica de Schwarzschild

$$g_{\mu\nu} = \begin{pmatrix} -\left(1-\frac{R}{r}\right) & 0 & 0 & 0 \\ 0 & \frac{1}{1-R/r} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2\sin^2\theta \end{pmatrix}$$



→ En relatividad general (RG) para una masa esférica:

El campo estacionario y simétrico viene dado por la métrica de Schwarzschild.

$$ds^2 = -\left(1-\frac{2GM}{c^2r}\right)c^2dt^2 + \left(1-\frac{2GM}{c^2r}\right)^{-1}dr^2 + r^2(d\theta^2 + \operatorname{Sen}^2\theta\, d\varphi^2)$$

Lagrangiano para una partícula de prueba:

sobre el plano $\theta = \pi/2$

$$2\mathcal{L} = g_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = -\left(1-\frac{2GM}{c^2r}\right)c^2\left(\frac{dt}{d\tau}\right)^2 + \left(1-\frac{2GM}{c^2r}\right)^{-1}\left(\frac{dr}{d\tau}\right)^2 + r^2\left(\frac{d\varphi}{d\tau}\right)^2$$

→ $\left(1-\frac{2GM}{c^2r}\right)c^2\left(\frac{dt}{d\tau}\right) = \tilde{E}$ (Conservación de energía)

$$\frac{dt}{d\tau} = \frac{\tilde{E}}{c^2\left(1-\frac{2GM}{c^2r}\right)}$$

→ $r^2\frac{d\varphi}{d\tau} = l \Rightarrow \frac{d\varphi}{d\tau} = \frac{\tilde{l}}{r^2}$ (Conservación del momento angular)

$\tilde{E}, \tilde{l}$, son de nuestra partícula de prueba.

2→ Ecuación radial (usando la normalización de la cuadrivelocidad)

Imponiendo $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -c^2$ $\quad \dot{} \equiv \frac{d}{d\tau}$

$$-\left(1-\frac{2GM}{c^2r}\right)c^2\left(\frac{dt}{d\tau}\right)^2 + \left(1-\frac{2GM}{c^2r}\right)^{-1}\left(\frac{dr}{d\tau}\right)^2 + r^2\left(\frac{d\varphi}{d\tau}\right)^2 = -c^2$$

↓ ↓ Reemplazando:

$$-\left(1-\frac{2GM}{c^2r}\right)c^2\left(\frac{\tilde{E}}{c^2\left(1-\frac{2GM}{c^2r}\right)}\right)^2 + \left(1-\frac{2GM}{c^2r}\right)^{-1}\dot{r}^2 + r^2\left(\frac{\tilde{l}}{r^2}\right)^2 = -c^2$$

$$\frac{-\tilde{E}^2}{c^2\left(1-\frac{2GM}{c^2r}\right)} + \left(1-\frac{2GM}{c^2r}\right)^{-1}\dot{r}^2 + \frac{\tilde{l}^2}{r^2} = -c^2$$

$$-\frac{\tilde{E}^2}{c^2} + \dot{r}^2 + \left(1-\frac{2GM}{c^2r}\right)\left(c^2+\frac{\tilde{l}^2}{r^2}\right) = 0$$

$$\dot{r}^2 = \frac{\tilde{E}^2}{c^2} - \left(1-\frac{2GM}{c^2r}\right)\left(c^2+\frac{\tilde{l}^2}{r^2}\right)$$

$$c^2 + \frac{l^2}{r^2} - \frac{2GM}{r} - \frac{2GM\tilde{l}^2}{cr^3}$$

$\Longrightarrow$ Buscamos:

$$\frac{1}{2}\dot{r}^2 + V_{eff}(r) = cste$$

$$\frac{1}{2}\dot{r}^2 = \underbrace{\frac{1}{2}\left(\frac{\tilde{E}^2}{c^2}-c^2\right)}_{cste} - \underbrace{\frac{1}{2}\left(\frac{\tilde{l}^2}{r^2}-\frac{2GM}{r}-\frac{2GM\tilde{l}^2}{c^2r^3}\right)}_{V_{eff}(r)}$$

2→ Cambio de Variable

$u(\varphi) = 1/r$ → ec. de la órbita $r(\varphi)$:

$$\frac{dr}{d\tau} = \frac{dr}{d\varphi}\frac{d\varphi}{d\tau}$$

$$\frac{dr}{d\tau} = -\frac{1}{u^2}\frac{du}{d\varphi}\,, \quad \frac{d\varphi}{d\tau} = \frac{\tilde{l}^2}{r^2} = \tilde{l}u^2$$

$$\Rightarrow \frac{dr}{d\tau} = -\frac{1}{u^2}\cdot\frac{du}{d\varphi}\cdot\tilde{l}u^2 = -\tilde{l}\frac{du}{d\varphi}$$

$$\text{o}_0\quad \dot{r}^2 = \tilde{l}^2\left(\frac{du}{d\varphi}\right)^2$$

$$\frac{1}{2}\dot{r}^2 = cste - \underbrace{\frac{1}{2}\left(\frac{\tilde{l}^2}{r^2}-\frac{2GM}{r}-\frac{2GM\tilde{l}^2}{c^2r^3}\right)}_{V_{eff}(r)}$$

$$\frac{d}{d\varphi}\left(\frac{1}{2}l^2\left(\frac{du}{d\varphi}\right)^2 = cste - \frac{1}{2}\left(u^2\tilde{l}^2 - 2GMu - \frac{2GM\tilde{l}^2}{c^2}\mu^3\right)\right)$$

$$\tilde{l}^2\left(\frac{du}{d\varphi}\right)\cdot\frac{d^2u}{d\varphi^2} = -\frac{1}{2}\left(2u\left(\frac{du}{d\varphi}\right)\tilde{l}^2 - 2GM\left(\frac{du}{d\varphi}\right) - 2GM\frac{\tilde{l}^2}{c^2}3\mu^2\left(\frac{du}{d\varphi}\right)\right)$$

$$\tilde{l}^2\frac{d^2u}{d\varphi^2} = -u\tilde{l}^2 + GM + 3\frac{GM\tilde{l}^2}{c^2}\mu^2$$

$$\boxed{\frac{d^2u}{d\varphi^2} + u = \frac{GM}{\tilde{l}^2} + 3\frac{GM}{c^2}u^2}$$

con $\mu = GM$

$$\frac{d^2u}{d\varphi^2} + u = \mu/\tilde{l}^2 + 3\mu/c^2\,u^2$$

ec. de mov. relativista:

$$\frac{d^2u}{d\varphi^2} + u = \frac{GM}{\tilde{l}^2} + 3\frac{GM}{c^2}u^2$$

con $\mu = GM$

$$\frac{d^2u}{d\varphi^2} + u = \mu/\tilde{l}^2 + 3\mu/c^2\, u^2$$

Con $c \to \infty$, recuperamos la ec. newtoniana:

$$\frac{d^2u}{d\varphi^2} + u = \frac{\mu}{\tilde{l}^2}$$

2. **(1p)** Derive la ecuación de movimiento que define la trayectoria de una onda gravitacional que se propaga en un espacio-tiempo curvo.

Ecuación de Einstein

$$G_{\mu\nu}[g] = 8\pi G T_{\mu\nu} \quad (\text{gravedad} = \text{Energía - Momento})$$

Partimos de un espacio-tiempo fondo $g^{(B)}_{\mu\nu}$ (que puede ser plano o curvo) y le agregamos una perturbación pequeña $h_{\mu\nu}$ que representa la onda gravitacional.

$$g_{\mu\nu} = g^{(B)}_{\mu\nu} + h_{\mu\nu}\,, \quad |h_{\mu\nu}| \ll 1$$

Desarrollo tipo Taylor

$$G_{\mu\nu}[\bar{g}+h] = G_{\mu\nu}[\bar{g}] + \delta G_{\mu\nu}[h] + \cancel{O(h^2)}$$

↑ fondo ↑ respuesta lineal a la perturbación

$$8\pi G\, T_{\mu\nu} = 8\pi G\, \bar{T}_{\mu\nu} + \delta G_{\mu\nu}[h]$$

$$\delta G_{\mu\nu}[h] = 8\pi G\,(T_{\mu\nu} - \bar{T}_{\mu\nu})$$

↳ Lo que gobierna la perturbación es el "exceso" de energía-momento sobre el fondo, en vacío de onda gravitacional $\delta G_{\mu\nu}[h] = 0$

↳ $\delta G_{\mu}[h] = \frac{1}{2}\left(-\bar{\Box} h_{\mu\nu} - \bar{\nabla}_\mu \bar{\nabla}_\nu h + \bar{\nabla}_\alpha \bar{\nabla}_\mu h^\alpha_\nu + \bar{\nabla}_\alpha \bar{\nabla}_\nu h^\alpha_\mu - \bar{g}_{\mu\nu} \bar{\nabla}_\alpha \bar{\nabla}_\beta h^{\alpha\beta} + \bar{g}_{\mu\nu} \bar{\Box} h\right)$

Definición de traza reducida:

donde $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\bar{g}_{\mu\nu} h$, con $h = \bar{g}^{\mu\nu} h_{\mu\nu}$

↳ $\delta G_{\mu\nu} = -\frac{1}{2}\left(\bar{\Box}\bar{h}_{\mu\nu} - \bar{\nabla}_\mu(\bar{\nabla}^\alpha \bar{h}_{\alpha\nu}) - \bar{\nabla}_\nu(\bar{\nabla}^\alpha \bar{h}_{\alpha\mu})\right) - \bar{R}_{\mu\alpha\nu\beta}\bar{h}^{\alpha\beta}$

↳ Gauge Armónico

$$\nabla^\mu \bar{h}_{\mu\nu} = 0$$

$$\bar{\Box}\bar{h}_{\mu\nu} + 2\bar{R}_{\mu\alpha\nu\beta}\bar{h}^{\alpha\beta} = -16\pi G(T_{\mu\nu} - \bar{T}_{\mu\nu})$$

De "onda" a "trayectoria": ansatz de alta frecuencia

$$h_{\mu\nu}(x) = A_{\mu\nu}(x)\, e^{i\theta(x)/\epsilon} \quad , \quad \epsilon \ll 1 \, , \quad K_\mu = \nabla_\mu \theta$$

(K_μ: vector propagación)

Orden dominante (la ecuación de dirección de la trayectoria)

$K^\mu K_\mu = 0$ → frente de onda es nula: se mueve como "luz" en el fondo

↳ ecuación dif:

$$u^\mu = \frac{dx^\mu}{d\lambda} = k^\mu \quad (\lambda \text{ como parámetro afín})$$

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\alpha\beta}(x)\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = 0$$ → Geodésica (no hay aceleración)

3. **(2p)** Derive la ecuación de movimiento de un electrón que se mueve en un espacio-tiempo curvo y en presencia de un campo electromagnético.

$$\mathcal{L} = -M\sqrt{-g_{\alpha\beta}(x)\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau}} + qA_\mu \dot{x}^\mu$$

(Energía cinética) (Potencial electromagnético)

$g_{\alpha\beta}(x)$ es la métrica del espacio-tiempo.

$x^\mu(\tau)$ es la trayectoria de la partícula.

τ es cualquier parámetro monótono ($\tau > 0$, no se detiene, no retrocede) y si es el tiempo propio el radical sería 1 ($\sqrt{g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu} = c$, $u^\mu u_u = c^2$)

4-velocidad siempre tiene norma c^2 cuando está parametrizada con tiempo propio.

$$S_{[x^\mu(\tau)]} = \int d\tau \left(-M\sqrt{-g_{\alpha\beta}(x)\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau}} + qA_\mu\dot{x}^\mu\right)$$

Es un funcional de las cuatro funciones $x^0(\tau), x^1(\tau), x^2(\tau), x^3(\tau)$

$$\delta S = 0$$ (S se minimiza con Euler-Lagrange)

$$\frac{d}{d\tau}\left(\frac{d\mathcal{L}}{d\dot{x}^\mu}\right) - \frac{d\mathcal{L}}{dx^\mu} = 0$$

$$\frac{d}{d\tau}\left(\frac{d}{d\dot{x}^\mu}\left(\mathcal{L}_K + \mathcal{L}_{EM}\right)\right) - \frac{d}{dx^\mu}\left(\mathcal{L}_K + \mathcal{L}_{EM}\right) = 0$$

$$\underbrace{\frac{d}{d\tau}\left(\frac{d\mathcal{L}_K}{d\dot{x}^\mu}\right) - \frac{d\mathcal{L}_K}{dx^\mu}}_{\text{solucionado en (4)}} + \frac{d}{d\tau}\left(\frac{d\mathcal{L}_{EM}}{d\dot{x}^\mu}\right) - \frac{d\mathcal{L}_{EM}}{dx^\mu} = 0$$

con: $\mathcal{L}_K = -M\sqrt{\underbrace{-g_{\alpha\beta}(x)\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau}}_{K}}$

$\mathcal{L}_K = -M\sqrt{K}$

$\mathcal{L}_{EM} = qA_\mu(x)\dot{x}^\mu$

$$M\left(\ddot{x}^\lambda + \Gamma^\lambda_{\sigma\nu}\dot{x}^\sigma\dot{x}^\nu\right) + \frac{d}{d\tau}\left(\frac{d\, qA_\nu(x)\dot{x}^\nu}{d\dot{x}^\mu}\right) - \frac{d\, qA_\nu(x)\dot{x}^\nu}{dx^\mu} = 0$$ ($\frac{d\dot{x}^\nu}{d\dot{x}^\mu} = \delta^\nu_\mu$)

$$\frac{d}{d\tau}\left(qA_\mu(x)\right) - q(\partial_\mu A_\nu(x))\dot{x}^\nu$$

$$q(\partial_\alpha A_\mu(x)\dot{x}^\alpha - (\partial_\mu A_\nu(x))\dot{x}^\nu) = 0 \quad \text{con } \nu = \alpha$$

$$\boxed{M\left(\ddot{x}^\lambda + \Gamma^\lambda_{\sigma\nu}\dot{x}^\sigma\dot{x}^\nu\right) = qF_{\mu\alpha}\dot{x}^\alpha}$$

4. **(3p)** Derive las ecuaciones de Euler-Lagrange de una acción cuya lagrangiana es

$$L = -M\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau}}.$$

$$\mathcal{L} = -M\sqrt{-g_{\alpha\beta}(x)\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau}}$$ (Lagrangiana estándar de una partícula libre)

$$S_{[x^\mu(\tau)]} = -M\int d\tau\sqrt{-g_{\alpha\beta}(x)\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau}}$$

$g_{\alpha\beta}(x)$ es la métrica del espacio-tiempo.

$x^\mu(\tau)$ es la trayectoria de la partícula.

τ es cualquier parámetro monótono ($\tau>0$, no se detiene, no retrocede) y si es el tiempo propio el radical sería 1 ($\sqrt{g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu}=c$, $u^\mu u_\mu = c^2$)

4-velocidad siempre tiene norma c^2 cuando está parametrizada con tiempo propio.

Es un funcional de las cuatro funciones $x^0(\tau), x^1(\tau), x^2(\tau), x^3(\tau)$

$$S:\{\text{curvas } x^\mu(\tau)\}\longrightarrow\mathbb{R}$$

$\delta S = 0$ (S se minimiza con Euler-Lagrange)

Ecuaciones de Euler-Lagrange

$$\frac{d}{d\tau}\left(\frac{d\mathcal{L}}{d\dot{x}^\mu}\right)-\frac{d\mathcal{L}}{dx^\mu}=0$$

con: $\mathcal{L} = -M\sqrt{\underbrace{-g_{\alpha\beta}(x)\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau}}_{K}}$

$\mathcal{L} = -M\sqrt{K}$

$$\frac{d}{d\tau}\left(\frac{d}{d\dot{x}^\mu}\left(-M\sqrt{K}\right)\right)-\frac{d}{dx^\mu}\left(-M\sqrt{K}\right)=\underbrace{\frac{d}{d\tau}\left(\frac{d}{d\dot{x}^\mu}\left(-MK^{1/2}\right)\right)}_{①}-\underbrace{\frac{d}{dx^\mu}\left(-MK^{1/2}\right)}_{②}=0$$

① $-\frac{1}{2}MK^{-1/2}\frac{dK}{d\dot{x}^\mu} \overset{*}{=} -\frac{1}{2}MK^{-1/2}\cdot -2\,g_{\mu\beta}(x)\dot{x}^\beta = MK^{-1/2}g_{\mu\beta}(x)\dot{x}^\beta$

$$\Rightarrow \frac{dK}{d\dot{x}^\mu} = \frac{d}{d\dot{x}^\mu}\left(-g_{\alpha\beta}(x)\,\dot{x}^\alpha\dot{x}^\beta\right)$$

$$\frac{dK}{d\dot{x}^\mu} = -g_{\alpha\beta}(x)\left(\frac{d\dot{x}^\alpha}{d\dot{x}^\mu}\dot{x}^\beta+\dot{x}^\alpha\frac{d\dot{x}^\beta}{d\dot{x}^\mu}\right)$$

$$= -g_{\alpha\beta}(x)\left(\delta^\alpha_\mu\dot{x}^\beta+\dot{x}^\alpha\delta^\beta_\mu\right)$$

$$= -g_{\mu\beta}(x)\dot{x}^\beta - \underbrace{g_{\alpha\mu}}_{g_{\mu\alpha}}(x)\dot{x}^\alpha$$ simetría y α es dummy $\Rightarrow \alpha=\beta$

$$\frac{dK}{d\dot{x}^\mu} = -g_{\mu\beta}(x)\dot{x}^\beta - g_{\mu\beta}(x)\dot{x}^\beta = -2\,g_{\mu\beta}(x)\dot{x}^\beta \quad (*)$$

$$\frac{d}{d\dot{x}^\mu}\left(\mathcal{L}\right) = M\frac{g_{\mu\beta}(x)\dot{x}^\beta}{\sqrt{-g_{\alpha\gamma}(x)\frac{dx^\alpha}{d\tau}\frac{dx^\gamma}{d\tau}}}$$

② $\frac{d}{d\tau^\mu}\left(-MK^{1/2}\right) = -\frac{1}{2}MK^{-1/2}\frac{dK}{dx^\mu}$

$$\frac{dK}{dx^\mu} = -\partial_\mu\, g_{\alpha\beta}(x)\,\dot{x}^\alpha\dot{x}^\beta$$

$$\frac{d}{d\tau^\mu}\left(-MK^{1/2}\right) = -\frac{1}{2}MK^{-1/2}\cdot -\partial_\mu\, g_{\alpha\beta}(x)\,\dot{x}^\alpha\dot{x}^\beta$$

$$\frac{d}{d\tau^\mu}\left(\mathcal{L}\right) = M\frac{\partial_\mu g_{\alpha\beta}(x)}{2\sqrt{K}}\dot{x}^\alpha\dot{x}^\beta$$

$$\Rightarrow \underbrace{\frac{d}{d\tau}\left(M\frac{g_{\mu\beta}(x)\dot{x}^\beta}{\sqrt{-g_{\alpha\gamma}(x)\frac{dx^\alpha}{d\tau}\frac{dx^\gamma}{d\tau}}}\right)}_{③} - M\frac{\partial_\mu g_{\alpha\beta}(x)}{2\sqrt{K}}\dot{x}^\alpha\dot{x}^\beta = 0$$

$$\Rightarrow \underbrace{\frac{d}{d\tau}\left(M \frac{g_{\mu\beta}(x)\,\dot{x}^{\beta}}{\sqrt{-g_{\alpha\beta}(x)\frac{dx^{\alpha}}{d\tau}\frac{dx^{\beta}}{d\tau}}} \right)}_{③} - M\frac{\partial_{\mu} g_{\alpha\beta}(x)}{2\sqrt{K}}\dot{x}^{\alpha}\dot{x}^{\beta} = 0$$

③ $$M\frac{d}{d\tau}\left(g_{\mu\beta}(x)\,\dot{x}^{\beta}\cdot\frac{1}{\sqrt{-g_{\alpha\beta}(x)\frac{dx^{\alpha}}{d\tau}\frac{dx^{\beta}}{d\tau}}} \right) = M\left(\frac{\frac{d}{d\tau}\left(g_{\mu\beta}(x)\,\dot{x}^{\beta}\right)}{\sqrt{-g_{\alpha\beta}(x)\frac{dx^{\alpha}}{d\tau}\frac{dx^{\beta}}{d\tau}}} + g_{\mu\beta}(x)\dot{x}^{\beta}\,\frac{d}{d\tau}\left(K^{-1/2}\right) \right)$$

$$= M\left(\frac{\frac{d}{d\tau}\left(g_{\mu\beta}(x)\right)\dot{x}^{\beta} + g_{\mu\beta}(x)\frac{d}{d\tau}\left(\dot{x}^{\beta}\right)}{\sqrt{-g_{\alpha\beta}(x)\frac{dx^{\alpha}}{d\tau}\frac{dx^{\beta}}{d\tau}}} + g_{\mu\beta}(x)\dot{x}^{\beta}\,\frac{d}{d\tau}\left(K^{-1/2}\right) \right)$$

$$= M\left(\frac{\frac{d}{d\tau}\left(g_{\mu\beta}(x)\right)\dot{x}^{\beta} + g_{\mu\beta}(x)\ddot{x}^{\beta}}{\sqrt{-g_{\alpha\beta}(x)\frac{dx^{\alpha}}{d\tau}\frac{dx^{\beta}}{d\tau}}} + g_{\mu\beta}(x)\dot{x}^{\beta}\,\frac{d}{d\tau}\left(K^{-1/2}\right) \right)$$

$$= M\left(\frac{\partial_{\delta} g_{\mu\beta}(x)\dot{x}^{\delta}\dot{x}^{\beta} + g_{\mu\beta}(x)\ddot{x}^{\beta}}{\sqrt{-g_{\alpha\beta}(x)\frac{dx^{\alpha}}{d\tau}\frac{dx^{\beta}}{d\tau}}} - \frac{1}{2} g_{\mu\beta}(x)\dot{x}^{\beta} K^{-3/2}\frac{dK}{d\tau} \right)$$

$$-\frac{d}{d\tau} g_{\alpha\beta}(x)\dot{x}^{\alpha}\dot{x}^{\beta} = -\partial_{\delta} g_{\alpha\beta}(x)\dot{x}^{\delta}\dot{x}^{\alpha}\dot{x}^{\beta} - g_{\alpha\beta}(x)\ddot{x}^{\alpha}\dot{x}^{\beta} - g_{\alpha\beta}(x)\dot{x}^{\alpha}\ddot{x}^{\beta}$$

$$= -\partial_{\delta} g_{\alpha\beta}(x)\dot{x}^{\delta}\dot{x}^{\alpha}\dot{x}^{\beta} - 2 g_{\alpha\beta}(x)\dot{x}^{\alpha}\ddot{x}^{\beta}$$

$$\Rightarrow M\left(\frac{\partial_{\delta} g_{\mu\beta}(x)\dot{x}^{\delta}\dot{x}^{\beta} + g_{\mu\beta}(x)\ddot{x}^{\beta}}{\sqrt{K}} - \frac{1}{2} g_{\mu\alpha}(x)\dot{x}^{\alpha} K^{-3/2}\left(\frac{dK}{d\tau}\right) \right) - M\frac{\partial_{\mu} g_{\alpha\beta}(x)}{2\sqrt{K}}\dot{x}^{\alpha}\dot{x}^{\beta} = 0$$

$$\partial_{\delta} g_{\mu\beta}(x)\dot{x}^{\delta}\dot{x}^{\beta} + g_{\mu\beta}(x)\ddot{x}^{\beta} - \frac{1}{2} g_{\mu\beta}(x)\dot{x}^{\beta} K^{-1}\left(\frac{dK}{d\tau}\right) - \frac{1}{2}\partial_{\mu} g_{\alpha\beta}(x)\dot{x}^{\alpha}\dot{x}^{\beta} = 0$$

$$\partial_{\delta} g_{\mu\beta}(x)\dot{x}^{\delta}\dot{x}^{\beta} + g_{\mu\beta}(x)\ddot{x}^{\beta} - \frac{1}{2} g_{\mu\beta}(x)\dot{x}^{\beta} K^{-1}\left(\frac{dK}{d\tau}\right) - \frac{1}{2}\partial_{\mu} g_{\alpha\beta}(x)\dot{x}^{\alpha}\dot{x}^{\beta} = 0$$

$$\left(\partial_{\alpha} g_{\mu\beta} - \tfrac{1}{2}\partial_{\mu} g_{\alpha\beta}\right)\dot{x}^{\alpha}\dot{x}^{\beta} + g_{\mu\beta}(x)\ddot{x}^{\beta} - \frac{1}{2} g_{\mu\beta}(x)\dot{x}^{\beta} K^{-1}\left(\frac{dK}{d\tau}\right) = 0$$

$\rightarrow \partial_{\sigma} g_{\mu\nu}\dot{x}^{\sigma}\dot{x}^{\nu}$

$\tfrac{1}{2}\left(\partial_{\sigma} g_{\mu\nu} + \partial_{\nu} g_{\mu\sigma}\right)\dot{x}^{\sigma}\dot{x}^{\nu}$

$$\tfrac{1}{2}\left(\partial_{\sigma} g_{\mu\nu} + \partial_{\nu} g_{\mu\sigma} - \partial_{\mu} g_{\sigma\nu}\right)\dot{x}^{\sigma}\dot{x}^{\nu} + g_{\mu\beta}(x)\ddot{x}^{\beta} - \frac{1}{2} g_{\mu\beta}(x)\dot{x}^{\beta} K^{-1}\left(\frac{dK}{d\tau}\right) = 0$$

$$\underbrace{\tfrac{1}{2} g^{\lambda\mu}\left(\partial_{\sigma} g_{\mu\nu} + \partial_{\nu} g_{\mu\sigma} - \partial_{\mu} g_{\sigma\nu}\right)}_{\Gamma^{\lambda}_{\sigma\nu}}\dot{x}^{\sigma}\dot{x}^{\nu} + \underbrace{g^{\lambda\mu} g_{\mu\beta}(x)}_{\delta^{\lambda}_{\beta}}\ddot{x}^{\beta} - \frac{1}{2}\underbrace{g^{\lambda\mu} g_{\mu\beta}(x)}_{\delta^{\lambda}_{\beta}}\dot{x}^{\beta} K^{-1}\left(\frac{dK}{d\tau}\right) = 0$$

$$\Gamma^{\lambda}_{\sigma\nu}\dot{x}^{\sigma}\dot{x}^{\nu} + \ddot{x}^{\lambda} - \frac{1}{2}\dot{x}^{\lambda} K^{-1}\left(\frac{dK}{d\tau}\right) = 0$$

$\frac{dK}{d\tau} = 0$ *

Ecuación de Geodesicas:

$$\ddot{x}^{\lambda} + \Gamma^{\lambda}_{\sigma\nu}\dot{x}^{\sigma}\dot{x}^{\nu} = 0$$

Símbolos de Christoffel

$$\Gamma^{\lambda}_{\sigma\nu} = \tfrac{1}{2} g^{\lambda\mu}\left(\partial_{\sigma} g_{\mu\nu} + \partial_{\nu} g_{\mu\sigma} - \partial_{\mu} g_{\sigma\nu}\right)$$

* Si elegimos τ afín, es decir, un parámetro para el cual K es cste a lo largo de la curva, entonces $\frac{dK}{d\tau} = 0$. Un caso natural es tomar τ como tiempo propio para una partícula timelike,

$$K(\text{Tiempo propio}) = c^2$$

5. **(3p)** Utilizando la definición de la derivada covariante, reduzca las siguientes expresiones:

   a) $(\nabla_\alpha\nabla_\beta - \nabla_\beta\nabla_\alpha)T^\mu$

   b) $(\nabla_\alpha\nabla_\beta - \nabla_\beta\nabla_\alpha)T_\mu$

   c) $(\nabla_\alpha\nabla_\beta - \nabla_\beta\nabla_\alpha)T^{\mu\nu}_{\rho\sigma}$

a) $[\nabla_\rho,\nabla_\sigma]V^\mu = (\nabla_\rho\nabla_\sigma - \nabla_\sigma\nabla_\rho)V^\mu$

$= \nabla_\rho\left(\partial_\sigma V^\mu + \Gamma^\mu_{\sigma\lambda}V^\lambda\right) - \nabla_\sigma\left(\partial_\rho V^\mu + \Gamma^\mu_{\rho\lambda}V^\lambda\right)$

$= \nabla_\rho\left(\partial_\sigma V^\mu\right) + \nabla_\rho\left(\Gamma^\mu_{\sigma\lambda}V^\lambda\right) - \nabla_\sigma\left(\partial_\rho V^\mu\right) - \nabla_\sigma\left(\Gamma^\mu_{\rho\lambda}V^\lambda\right)$

$= \nabla_\rho\left(\partial_\sigma V^\mu\right) + \Gamma^\mu_{\sigma\lambda}\nabla_\rho\left(V^\lambda\right) + \nabla_\rho\left(\Gamma^\mu_{\sigma\lambda}\right)V^\lambda - \nabla_\rho\left(\partial_\sigma V^\mu\right) - \Gamma^\mu_{\sigma\lambda}\nabla_\rho\left(V^\lambda\right) - \nabla_\rho\left(\Gamma^\mu_{\sigma\lambda}\right)V^\lambda$

$= \partial_\rho\partial_\sigma V^\mu + \Gamma^\mu_{\rho\alpha}\left(\partial_\sigma V^\alpha\right) - \Gamma^\alpha_{\rho\sigma}\left(\partial_\alpha V^\mu\right) + \Gamma^\mu_{\sigma\lambda}\partial_\rho V^\lambda + \Gamma^\mu_{\sigma\lambda}\Gamma^\lambda_{\rho\alpha}V^\alpha + \partial_\rho\Gamma^\mu_{\sigma\lambda}V^\lambda + \Gamma^\mu_{\rho\eta}\Gamma^\eta_{\sigma\lambda}V^\lambda - \Gamma^\eta_{\rho\sigma}\Gamma^\mu_{\eta\lambda}V^\lambda - \Gamma^\eta_{\rho\lambda}\Gamma^\mu_{\sigma\eta}V^\lambda$

$-\left(\partial_\sigma\partial_\rho V^\mu + \Gamma^\mu_{\sigma\alpha}\left(\partial_\rho V^\alpha\right) - \Gamma^\alpha_{\sigma\rho}\left(\partial_\alpha V^\mu\right) + \Gamma^\mu_{\rho\lambda}\partial_\sigma V^\lambda + \Gamma^\mu_{\rho\lambda}\Gamma^\lambda_{\sigma\alpha}V^\alpha + \partial_\sigma\Gamma^\mu_{\rho\lambda}V^\lambda + \Gamma^\mu_{\sigma\eta}\Gamma^\eta_{\rho\lambda}V^\lambda - \Gamma^\eta_{\sigma\rho}\Gamma^\mu_{\eta\lambda}V^\lambda - \Gamma^\eta_{\sigma\lambda}\Gamma^\mu_{\rho\eta}V^\lambda\right)$

$[\nabla_\rho,\nabla_\sigma]V^\mu = \left(\partial_\rho\Gamma^\mu_{\sigma\alpha} - \partial_\sigma\Gamma^\mu_{\rho\alpha} + \Gamma^\mu_{\sigma\lambda}\Gamma^\lambda_{\rho\alpha} - \Gamma^\mu_{\rho\lambda}\Gamma^\lambda_{\sigma\alpha}\right)V^\alpha$

$R^\mu_{\alpha\rho\sigma} = \partial_\rho\Gamma^\mu_{\alpha\sigma} + \Gamma^\mu_{\lambda\sigma}\Gamma^\lambda_{\alpha\rho} - \partial_\sigma\Gamma^\mu_{\alpha\rho} - \Gamma^\mu_{\lambda\rho}\Gamma^\lambda_{\alpha\sigma}$

operador (tensor de Riemann)

$$\boxed{[\nabla_\rho,\nabla_\sigma]V^\mu = R^\mu_{\alpha\rho\sigma}V^\alpha}$$

b) $[\nabla_\rho,\nabla_\sigma]V_\mu = (\nabla_\rho\nabla_\sigma - \nabla_\sigma\nabla_\rho)V_\mu$

$V_\mu = g_{\mu\nu}V^\nu$

$[\nabla_\rho,\nabla_\sigma]V_\mu = [\nabla_\rho,\nabla_\sigma]\left(g_{\mu\nu}V^\nu\right) \quad (\nabla g = 0)$

$= g_{\mu\nu}[\nabla_\rho,\nabla_\sigma]V^\nu = g_{\mu\nu}R^\nu_{\alpha\rho\sigma}V^\alpha$

$[\nabla_\rho,\nabla_\sigma]V_\mu = R_{\mu\alpha\rho\sigma}V^\alpha$ ↳ $V^\alpha = g^{\alpha l}V_l$

$[\nabla_\rho,\nabla_\sigma]V_\mu = R_{\mu\alpha\rho\sigma}g^{\alpha l}V_l$

$[\nabla_\rho,\nabla_\sigma]V_\mu = R_{\mu}{}^{l}{}_{\rho\sigma}V_l$

cambiamos índices, R es asimétrico.

$$\boxed{[\nabla_\rho,\nabla_\sigma]V_\mu = -R^l{}_{\mu\rho\sigma}V_l}$$

c) $[\nabla_\rho, \nabla_\tau] V^{\mu\nu}{}_{\lambda\sigma} = (\nabla_\rho \nabla_\tau - \nabla_\tau \nabla_\rho) V^{\mu\nu}{}_{\lambda\sigma}$

$$\nabla_\rho V^{\mu\nu}{}_{\lambda\sigma} = \partial_\rho V^{\mu\nu}{}_{\lambda\sigma} + \underbrace{\Gamma^\mu_{\rho\alpha} V^{\alpha\nu}{}_{\lambda\sigma}}_{\mu} + \underbrace{\Gamma^\nu_{\rho\alpha} V^{\mu\alpha}{}_{\lambda\sigma}}_{\nu} - \underbrace{\Gamma^\alpha_{\rho\lambda} V^{\mu\nu}{}_{\alpha\sigma}}_{\lambda} - \underbrace{\Gamma^\alpha_{\rho\sigma} V^{\mu\nu}{}_{\lambda\alpha}}_{\sigma}$$

$$\nabla_\tau V^{\mu\nu}{}_{\lambda\sigma} = \partial_\tau V^{\mu\nu}{}_{\lambda\sigma} + \underbrace{\Gamma^\mu_{\tau\alpha} V^{\alpha\nu}{}_{\lambda\sigma}}_{\mu} + \underbrace{\Gamma^\nu_{\tau\alpha} V^{\mu\alpha}{}_{\lambda\sigma}}_{\nu} - \underbrace{\Gamma^\alpha_{\tau\lambda} V^{\mu\nu}{}_{\alpha\sigma}}_{\lambda} - \underbrace{\Gamma^\alpha_{\tau\sigma} V^{\mu\nu}{}_{\lambda\alpha}}_{\sigma}$$

$$[\nabla_\rho, \nabla_\tau] V^{\mu\nu}{}_{\lambda\sigma} = \nabla_\rho (\nabla_\tau V^{\mu\nu}{}_{\lambda\sigma}) - \nabla_\tau (\nabla_\rho V^{\mu\nu}{}_{\lambda\sigma})$$

$$= \nabla_\rho \left(\partial_\tau V^{\mu\nu}{}_{\lambda\sigma} + \Gamma^\mu_{\tau\alpha} V^{\alpha\nu}{}_{\lambda\sigma} + \Gamma^\nu_{\tau\alpha} V^{\mu\alpha}{}_{\lambda\sigma} - \Gamma^\alpha_{\tau\lambda} V^{\mu\nu}{}_{\alpha\sigma} - \Gamma^\alpha_{\tau\sigma} V^{\mu\nu}{}_{\lambda\alpha} \right) - \nabla_\tau \left(\partial_\rho V^{\mu\nu}{}_{\lambda\sigma} + \Gamma^\mu_{\rho\alpha} V^{\alpha\nu}{}_{\lambda\sigma} + \Gamma^\nu_{\rho\alpha} V^{\mu\alpha}{}_{\lambda\sigma} - \Gamma^\alpha_{\rho\lambda} V^{\mu\nu}{}_{\alpha\sigma} - \Gamma^\alpha_{\rho\sigma} V^{\mu\nu}{}_{\lambda\alpha} \right)$$

$$= \underbrace{\nabla_\rho (\partial_\tau V^{\mu\nu}{}_{\lambda\sigma})}_{①} + \underbrace{\Gamma^\mu_{\tau\alpha} \nabla_\rho (V^{\alpha\nu}{}_{\lambda\sigma})}_{②} + \nabla_\rho (\Gamma^\mu_{\tau\alpha}) V^{\alpha\nu}{}_{\lambda\sigma} + \underbrace{\Gamma^\nu_{\tau\alpha} \nabla_\rho (V^{\mu\alpha}{}_{\lambda\sigma})}_{③} + \nabla_\rho (\Gamma^\nu_{\tau\alpha}) V^{\mu\alpha}{}_{\lambda\sigma} \underbrace{- \Gamma^\alpha_{\tau\lambda} \nabla_\rho (V^{\mu\nu}{}_{\alpha\sigma})}_{④} - \nabla_\rho (\Gamma^\alpha_{\tau\lambda}) V^{\mu\nu}{}_{\alpha\sigma} \underbrace{- \Gamma^\alpha_{\tau\sigma} \nabla_\rho (V^{\mu\nu}{}_{\lambda\alpha})}_{⑤} - \nabla_\rho (\Gamma^\alpha_{\tau\sigma}) V^{\mu\nu}{}_{\lambda\alpha} -$$

$$\underbrace{- \nabla_\tau (\partial_\rho V^{\mu\nu}{}_{\lambda\sigma})}_{⑥} \underbrace{- \Gamma^\mu_{\rho\alpha} \nabla_\tau (V^{\alpha\nu}{}_{\lambda\sigma})}_{⑦} - \nabla_\tau (\Gamma^\mu_{\rho\alpha}) V^{\alpha\nu}{}_{\lambda\sigma} \underbrace{- \Gamma^\nu_{\rho\alpha} \nabla_\tau (V^{\mu\alpha}{}_{\lambda\sigma})}_{⑧} - \nabla_\tau (\Gamma^\nu_{\rho\alpha}) V^{\mu\alpha}{}_{\lambda\sigma} \underbrace{+ \Gamma^\alpha_{\rho\lambda} \nabla_\tau (V^{\mu\nu}{}_{\alpha\sigma})}_{⑨} + \nabla_\tau (\Gamma^\alpha_{\rho\lambda}) V^{\mu\nu}{}_{\alpha\sigma} \underbrace{+ \Gamma^\alpha_{\rho\sigma} \nabla_\tau (V^{\mu\nu}{}_{\lambda\alpha})}_{⑩} + \nabla_\tau (\Gamma^\alpha_{\rho\sigma}) V^{\mu\nu}{}_{\lambda\alpha}$$

① $\nabla_\rho (\partial_\tau V^{\mu\nu}{}_{\lambda\sigma}) = \partial_\rho (\partial_\tau V^{\mu\nu}{}_{\lambda\sigma}) + \Gamma^\mu_{\rho\alpha} (\partial_\tau V^{\alpha\nu}{}_{\lambda\sigma}) + \Gamma^\nu_{\rho\alpha} (\partial_\tau V^{\mu\alpha}{}_{\lambda\sigma}) - \Gamma^\alpha_{\rho\lambda} (\partial_\tau V^{\mu\nu}{}_{\alpha\sigma}) - \Gamma^\alpha_{\rho\sigma} (\partial_\tau V^{\mu\nu}{}_{\lambda\alpha})$

② $\Gamma^\mu_{\tau\alpha} \nabla_\rho (V^{\alpha\nu}{}_{\lambda\sigma}) = \Gamma^\mu_{\tau\alpha} \partial_\rho (V^{\alpha\nu}{}_{\lambda\sigma}) + \Gamma^\mu_{\tau\alpha} \Gamma^\alpha_{\rho\beta} V^{\beta\nu}{}_{\lambda\sigma} + \Gamma^\mu_{\tau\alpha} \Gamma^\nu_{\rho\beta} V^{\alpha\beta}{}_{\lambda\sigma} - \Gamma^\mu_{\tau\alpha} \Gamma^\beta_{\rho\lambda} V^{\alpha\nu}{}_{\beta\sigma} - \Gamma^\mu_{\tau\alpha} \Gamma^\beta_{\rho\sigma} V^{\alpha\nu}{}_{\lambda\beta}$

③ $\Gamma^\nu_{\tau\alpha} \nabla_\rho (V^{\mu\alpha}{}_{\lambda\sigma}) = \Gamma^\nu_{\tau\alpha} \partial_\rho (V^{\mu\alpha}{}_{\lambda\sigma}) + \Gamma^\nu_{\tau\alpha} \Gamma^\mu_{\rho\beta} V^{\beta\alpha}{}_{\lambda\sigma} + \Gamma^\nu_{\tau\alpha} \Gamma^\alpha_{\rho\beta} V^{\mu\beta}{}_{\lambda\sigma} - \Gamma^\nu_{\tau\alpha} \Gamma^\beta_{\rho\lambda} V^{\mu\alpha}{}_{\beta\sigma} - \Gamma^\nu_{\tau\alpha} \Gamma^\beta_{\rho\sigma} V^{\mu\alpha}{}_{\lambda\beta}$

④ $\Gamma^\alpha_{\tau\lambda} \nabla_\rho (V^{\mu\nu}{}_{\alpha\sigma}) = \Gamma^\alpha_{\tau\lambda} \partial_\rho (V^{\mu\nu}{}_{\alpha\sigma}) + \Gamma^\alpha_{\tau\lambda} \Gamma^\mu_{\rho\beta} V^{\beta\nu}{}_{\alpha\sigma} + \Gamma^\alpha_{\tau\lambda} \Gamma^\nu_{\rho\beta} V^{\mu\beta}{}_{\alpha\sigma} - \Gamma^\alpha_{\tau\lambda} \Gamma^\beta_{\rho\alpha} V^{\mu\nu}{}_{\beta\sigma} - \Gamma^\alpha_{\tau\lambda} \Gamma^\beta_{\rho\sigma} V^{\mu\nu}{}_{\alpha\beta}$ (−)

⑤ $\Gamma^\alpha_{\tau\sigma} \nabla_\rho (V^{\mu\nu}{}_{\lambda\alpha}) = \Gamma^\alpha_{\tau\sigma} \partial_\rho (V^{\mu\nu}{}_{\lambda\alpha}) + \Gamma^\alpha_{\tau\sigma} \Gamma^\mu_{\rho\beta} V^{\beta\nu}{}_{\lambda\alpha} + \Gamma^\alpha_{\tau\sigma} \Gamma^\nu_{\rho\beta} V^{\mu\beta}{}_{\lambda\alpha} - \Gamma^\alpha_{\tau\sigma} \Gamma^\beta_{\rho\lambda} V^{\mu\nu}{}_{\beta\alpha} - \Gamma^\alpha_{\tau\sigma} \Gamma^\beta_{\rho\alpha} V^{\mu\nu}{}_{\lambda\beta}$ (−)

⑥ $\nabla_\tau (\partial_\rho V^{\mu\nu}{}_{\lambda\sigma}) = \partial_\tau (\partial_\rho V^{\mu\nu}{}_{\lambda\sigma}) + \Gamma^\mu_{\tau\alpha} (\partial_\rho V^{\alpha\nu}{}_{\lambda\sigma}) + \Gamma^\nu_{\tau\alpha} (\partial_\rho V^{\mu\alpha}{}_{\lambda\sigma}) - \Gamma^\alpha_{\tau\lambda} (\partial_\rho V^{\mu\nu}{}_{\alpha\sigma}) - \Gamma^\alpha_{\tau\sigma} (\partial_\rho V^{\mu\nu}{}_{\lambda\alpha})$ (−)

⑦ $\Gamma^\mu_{\rho\alpha} \nabla_\tau (V^{\alpha\nu}{}_{\lambda\sigma}) = \Gamma^\mu_{\rho\alpha} \partial_\tau (V^{\alpha\nu}{}_{\lambda\sigma}) + \Gamma^\mu_{\rho\alpha} \Gamma^\alpha_{\tau\beta} V^{\beta\nu}{}_{\lambda\sigma} + \Gamma^\mu_{\rho\alpha} \Gamma^\nu_{\tau\beta} V^{\alpha\beta}{}_{\lambda\sigma} - \Gamma^\mu_{\rho\alpha} \Gamma^\beta_{\tau\lambda} V^{\alpha\nu}{}_{\beta\sigma} - \Gamma^\mu_{\rho\alpha} \Gamma^\beta_{\tau\sigma} V^{\alpha\nu}{}_{\lambda\beta}$ (−)

⑧ $\Gamma^\nu_{\rho\alpha} \nabla_\tau (V^{\mu\alpha}{}_{\lambda\sigma}) = \Gamma^\nu_{\rho\alpha} \partial_\tau (V^{\mu\alpha}{}_{\lambda\sigma}) + \Gamma^\nu_{\rho\alpha} \Gamma^\mu_{\tau\beta} V^{\beta\alpha}{}_{\lambda\sigma} + \Gamma^\nu_{\rho\alpha} \Gamma^\alpha_{\tau\beta} V^{\mu\beta}{}_{\lambda\sigma} - \Gamma^\nu_{\rho\alpha} \Gamma^\beta_{\tau\lambda} V^{\mu\alpha}{}_{\beta\sigma} - \Gamma^\nu_{\rho\alpha} \Gamma^\beta_{\tau\sigma} V^{\mu\alpha}{}_{\lambda\beta}$ (−)

⑨ $\Gamma^\alpha_{\rho\lambda} \nabla_\tau (V^{\mu\nu}{}_{\alpha\sigma}) = \Gamma^\alpha_{\rho\lambda} \partial_\tau (V^{\mu\nu}{}_{\alpha\sigma}) + \Gamma^\alpha_{\rho\lambda} \Gamma^\mu_{\tau\beta} V^{\beta\nu}{}_{\alpha\sigma} + \Gamma^\alpha_{\rho\lambda} \Gamma^\nu_{\tau\beta} V^{\mu\beta}{}_{\alpha\sigma} - \Gamma^\alpha_{\rho\lambda} \Gamma^\beta_{\tau\alpha} V^{\mu\nu}{}_{\beta\sigma} - \Gamma^\alpha_{\rho\lambda} \Gamma^\beta_{\tau\sigma} V^{\mu\nu}{}_{\alpha\beta}$ (+)

⑩ $\Gamma^\alpha_{\rho\sigma} \nabla_\tau (V^{\mu\nu}{}_{\lambda\alpha}) = \Gamma^\alpha_{\rho\sigma} \partial_\tau (V^{\mu\nu}{}_{\lambda\alpha}) + \Gamma^\alpha_{\rho\sigma} \Gamma^\mu_{\tau\beta} V^{\beta\nu}{}_{\lambda\alpha} + \Gamma^\alpha_{\rho\sigma} \Gamma^\nu_{\tau\beta} V^{\mu\beta}{}_{\lambda\alpha} - \Gamma^\alpha_{\rho\sigma} \Gamma^\beta_{\tau\lambda} V^{\mu\nu}{}_{\beta\alpha} - \Gamma^\alpha_{\rho\sigma} \Gamma^\beta_{\tau\alpha} V^{\mu\nu}{}_{\lambda\beta}$ (+)

$\Gamma^\mu_{\tau\alpha} \Gamma^\alpha_{\rho\beta} V^{\beta\nu}{}_{\lambda\sigma} + \nabla_\rho (\Gamma^\mu_{\tau\alpha}) V^{\alpha\nu}{}_{\lambda\sigma} - \nabla_\tau (\Gamma^\mu_{\rho\alpha}) V^{\alpha\nu}{}_{\lambda\sigma} - \Gamma^\mu_{\rho\alpha} \Gamma^\alpha_{\tau\beta} V^{\beta\nu}{}_{\lambda\sigma} \rightsquigarrow R^\mu{}_{\beta\rho\tau} V^{\beta\nu}{}_{\lambda\sigma}$

$-\Gamma^\nu_{\tau\alpha} \Gamma^\mu_{\rho\beta} V^{\beta\alpha}{}_{\lambda\sigma} - \nabla_\rho (\Gamma^\alpha_{\tau\lambda}) V^{\mu\nu}{}_{\alpha\sigma} + \nabla_\tau (\Gamma^\alpha_{\rho\lambda}) V^{\mu\nu}{}_{\alpha\sigma} + \Gamma^\alpha_{\rho\lambda} \Gamma^\mu_{\tau\beta} V^{\beta\nu}{}_{\alpha\sigma} \rightsquigarrow -R^\alpha{}_{\lambda\rho\tau} V^{\mu\nu}{}_{\alpha\sigma}$

$-\Gamma^\alpha_{\rho\lambda} \Gamma^\mu_{\tau\beta} V^{\beta\nu}{}_{\alpha\sigma} - \nabla_\rho (\Gamma^\nu_{\tau\alpha}) V^{\mu\alpha}{}_{\lambda\sigma} + \nabla_\tau (\Gamma^\nu_{\rho\alpha}) V^{\mu\alpha}{}_{\lambda\sigma} + \Gamma^\alpha_{\rho\lambda} \Gamma^\nu_{\tau\beta} V^{\mu\beta}{}_{\alpha\sigma} \rightsquigarrow -R^\nu{}_{\beta\rho\tau} V^{\mu\beta}{}_{\lambda\sigma}$

$\Gamma^\alpha_{\tau\beta} \Gamma^\mu_{\rho\beta} V^{\beta\nu}{}_{\lambda\alpha} + \nabla_\tau (\Gamma^\alpha_{\rho\lambda}) V^{\mu\nu}{}_{\alpha\sigma} - \nabla_\rho (\Gamma^\alpha_{\tau\beta}) V^{\mu\nu}{}_{\lambda\alpha} - \Gamma^\alpha_{\rho\beta} \Gamma^\mu_{\tau\beta} V^{\beta\nu}{}_{\lambda\alpha} \rightsquigarrow R^\mu{}_{\beta\rho\tau} V^{\beta\nu}{}_{\lambda\alpha}$

$$\boxed{[\nabla_\rho, \nabla_\tau] V^{\mu\nu}{}_{\lambda\sigma} = R^\mu{}_{\beta\rho\tau} V^{\beta\nu}{}_{\lambda\sigma} + R^\nu{}_{\beta\rho\tau} V^{\mu\beta}{}_{\lambda\sigma} - R^\alpha{}_{\lambda\rho\tau} V^{\mu\nu}{}_{\alpha\sigma} - R^\alpha{}_{\sigma\rho\tau} V^{\mu\nu}{}_{\lambda\alpha}}$$

6. **(2p)** Derive la ecuación del desvío geodésico y como aplicación explique lo que le sucede a un astronauta cerca al horizonte de un agujero negro.

Congruencia de geodésicas temporales $x^\mu(\tau, s)$

$$\Rightarrow T^\mu = \frac{\partial x^\mu}{\partial \tau}, \quad S^\mu = \frac{\partial x^\mu}{\partial s}$$

$$\cdot \frac{\partial}{\partial s}\frac{\partial x^\mu}{\partial \tau} = \frac{\partial}{\partial \tau}\frac{\partial x^\mu}{\partial s} \longrightarrow \frac{\partial T^\mu}{\partial s} = \frac{\partial S^\mu}{\partial \tau}$$

$$\frac{DS^\mu}{d\tau} = T^\nu \nabla_\nu S^\mu$$

$$T^\nu \nabla_\nu (S^\rho) - S^\nu \nabla_\nu (T^\rho) = [T, S]^\rho = 0$$

$[\partial\tau, \partial s] = 0$, T, S son campos coordenados

$$T^\nu \nabla_\nu S^\mu = S^\mu \nabla_\nu T^\mu$$

$$\frac{D^2 S^\mu}{d\tau^2} = T^\alpha \nabla_\alpha (S^\nu \nabla_\nu T^\mu)$$

$$= (T^\alpha \nabla_\alpha S^\nu)(\nabla_\nu T^\mu) + S^\nu T^\alpha \nabla_\alpha \nabla_\nu T^\mu$$

$$T^\alpha \nabla_\alpha S^\nu = S^\alpha \nabla_\alpha T^\nu$$

$$= (S^\alpha \nabla_\alpha T^\nu)(\nabla_\nu T^\mu) + S^\nu T^\alpha \nabla_\alpha \nabla_\nu T^\mu$$

$$T^\alpha \nabla_\alpha \nabla_\nu T^\mu = T^\alpha \nabla_\nu \nabla_\alpha T^\mu + T^\alpha R^\mu{}_{\lambda\alpha\nu}$$

$$\longrightarrow \nabla_\nu (T^\alpha \nabla_\alpha T^\mu) = \nabla_\nu (0) = 0$$

$$\nabla_\nu (T^\alpha \nabla_\alpha T^\mu) = (\nabla_\nu T^\alpha)(\nabla_\alpha T^\mu) + T^\alpha \nabla_\nu \nabla_\alpha T^\mu = 0$$

$$T^\alpha \nabla_\nu \nabla_\alpha T^\mu = -(\nabla_\nu T^\alpha)(\nabla_\alpha T^\mu)$$

$$T^\alpha \nabla_\alpha \nabla_\nu T^\mu = -(\nabla_\nu T^\alpha)(\nabla_\alpha T^\mu) + R^\mu{}_{\lambda\alpha\nu} T^\alpha T^\lambda$$

$$\frac{D^2 S^\mu}{d\tau^2} = (S^\alpha \nabla_\alpha T^\nu)(\nabla_\nu T^\mu) + S^\nu [-(\nabla_\nu T^\alpha)(\nabla_\alpha T^\mu + R^\mu{}_{\lambda\alpha\nu} T^\alpha T^\lambda]$$

$$\frac{D^2 S^\mu}{d\tau^2} = R^\mu{}_{\lambda\alpha\nu} T^\lambda T^\alpha S^\nu$$

$$R^\mu{}_{\lambda\alpha\nu} = -R^\mu{}_{\lambda\nu\alpha}$$

$$\frac{D^2 S^\mu}{d\tau^2} + R^\mu{}_{\nu\lambda\rho} T^\nu T^\lambda S^\rho = 0$$

o) la ecuación de desvio geodésica

$$\frac{D^2 S^\mu}{d\tau^2} + R^\mu{}_{\nu\lambda\rho} T^\nu T^\lambda S^\rho = 0$$

describe la aceleración relativa entre dos partículas en caída libre separadas por el vector S^μ.

Para un astronauta cerca del horizonte de un agujero negro de Schwarzschild, las componentes no nulas del tensor de Rieman en su sistema de referencia propio son

$$R^{\hat{r}}{}_{0\hat{r}0} = \frac{2GM}{r^3}, \quad R^{\theta}{}_{0\hat{\theta}0} = R^{\phi}{}_{0\hat{\phi}0} = -\frac{GM}{r^3}$$

$$\longrightarrow \frac{d^2 s^{\hat{r}}}{d\tau^2} = \frac{2GM}{r^3} s^{\hat{r}}, \quad \frac{d^2 s^\theta}{d\tau^2} = -\frac{GM}{r^3} s^\theta$$

$$r = r_s = 2GM/c^2$$

La magnitud de estas fuerzas son $\left.\frac{d^2 s^{\hat{r}}}{d\tau^2}\right|_{r_s} \approx \frac{c^6}{4G^2M^2} s^{\hat{r}}$

Para un agujero negro supermasivo ($M >> M_\odot$), este valor es pequeño y el astronauta cruza el horizonte sin daño inmediato. Para un agujero negro estelar ($M \sim M_\odot$), el término $c^6/(G^2M^2)$ es enorme, produciendo fuerzas destructivas antes de llegar a r_s.

Dentro del horizonte ($r < r_s$), las fuerzas crecen como $1/r^3$, llevando a la ESPAGUETIZACIÓN (estiramiento radial divergente y compresión lateral creciente) en un tiempo propio finito antes de alcanzar la singularidad en $r = 0$.

7. **(1p)** Redacte un resumen de la sección **9.2 Mach's Principle** (pp. 121-125) del libro *Ray D'Inverno, Introducing Einstein's Relativity.*

Aborda el principio como una idea física profunda que, aunque influyó decisivamente en Einstein, no se realiza plenamente en la estructura matemática de la relatividad general. La inercia de un cuerpo no es una propiedad intrínseca, sino consecuencia de su interacción con la distribución total de materia en el universo. Formalmente, esto sugiere que el tensor métrico $g_{\mu\nu}(x)$ debería estar enteramente determinado por el tensor energía impulso $T_{\mu\nu}(x)$ de toda materia cósmica

$$g_{\mu\nu}(x) = \mathcal{F}[T_{\mu\nu}(x')]$$

Incompatibilidad con la RG: Las ecuaciones de Einstein $G_{\mu\nu} = \kappa T_{\mu\nu}$ admiten soluciones de vacío no triviales ($T_{\mu\nu} = 0$) con geometría no plana, contradiciendo la exigencia de que "sin materia no hay geometría". Además, para resolverlas se requieren condiciones de contorno asintóticas, las cuales reintroducen una estructura de fondo independiente de la materia, justo lo que se quería eliminar.

El principio de Mach fue una guía valiosa, pero la relatividad general lo transciende al admitir geometrías no determinadas localmente por la materia. La sección subraya así la tensión irresoluble entre el ideal filosófico de Mach y el formalismo einsteiniano, concluyendo que el principio no es un teorema de la teoría, sino una motivación histórica que la teoría misma supera.